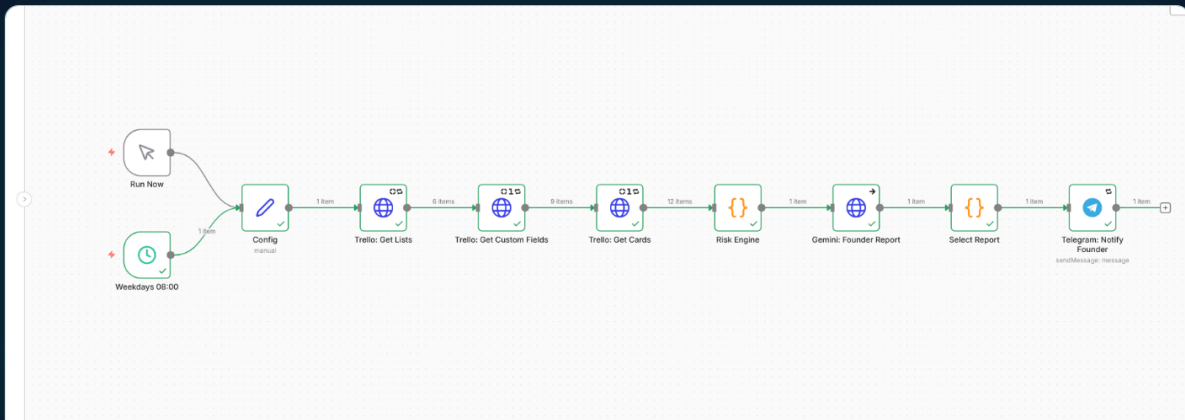


AI Project Manager

Trello + n8n + Gemini + Telegram

Reads every project on a Trello board, decides health with clear rules, has Gemini write a short founder report, checks it, and sends it on Telegram.

• OVERDUE • BLOCKED • AT RISK • ON TRACK



P08 • AI Project Manager • Trello + n8n + Gemini + Telegram • Muhammad Rehan Ansar

2026

CASE STUDY

Daily project health for a founder, from Trello to Telegram, with AI that can't overrule the facts

An n8n workflow reads a Trello board every weekday morning, decides each project's health with deterministic rules, asks Gemini to turn the results into a short founder note, checks that note against the rules, and delivers it on Telegram. If the AI fails or gets something wrong, the rule-based report is sent instead.

The data is in Trello, but the picture isn't

Teams already track delivery on a Trello board. The founder needs a quick answer to "what is late, what is stuck, and what needs me?" — but the signals are spread across due dates, custom fields, activity dates and links between cards. Checking by hand takes time, so problems are often noticed late.

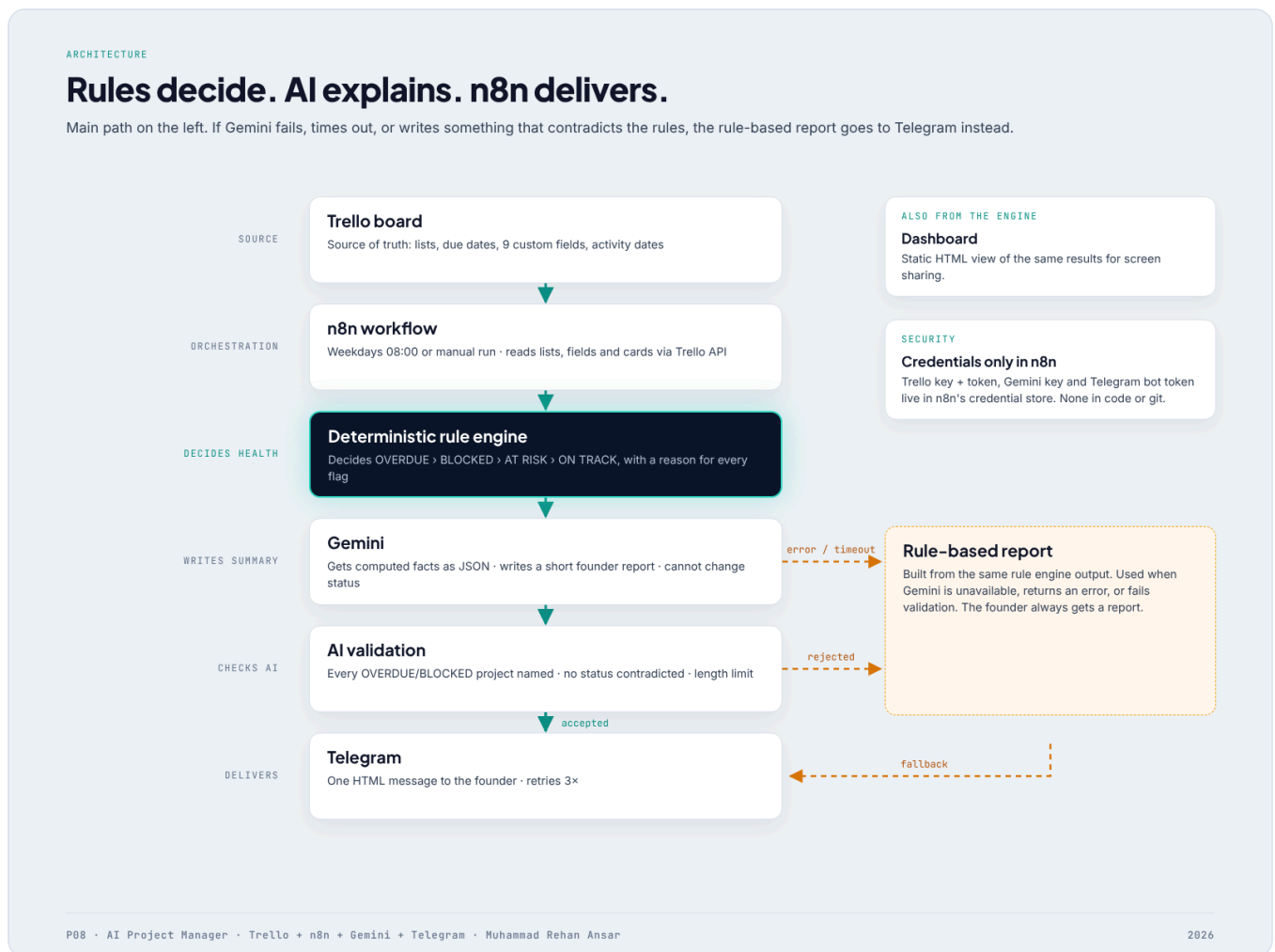
Simply asking an AI model to judge each project does not fix this: the answer can change from run to run, and nobody can see why a project was marked at risk.

Rules decide. AI explains. n8n delivers.

- **Trello** stays the single source of truth. No new tool for the team.
- **n8n** runs the flow on a weekday 08:00 schedule, or on demand.
- A **deterministic rule engine** sets OVERDUE, BLOCKED, AT RISK or ON TRACK, and records a reason for every flag.
- **Gemini** only writes the summary from those facts; it cannot change a status.
- A **validation step** rejects AI text that drops a critical project or contradicts the rules.
- **Telegram** delivers one clear message. The founder always gets a correct report.

LAYER	TECHNOLOGY	NOTES
Source	Trello REST API	Lists, custom fields, open cards · key + token via n8n Trello credential
Orchestration	n8n 1.123	Schedule + manual triggers, HTTP Request, Code and Telegram nodes
Rules	JavaScript (no dependencies)	One file, embedded unchanged in the n8n Code nodes, unit tested
AI	Google Gemini API	generateContent, temperature 0.2, 30 s timeout, errors continue the flow
Delivery	Telegram Bot API	HTML message, 3 retries
Testing	Node.js test runner	34 tests, 12 acceptance scenarios

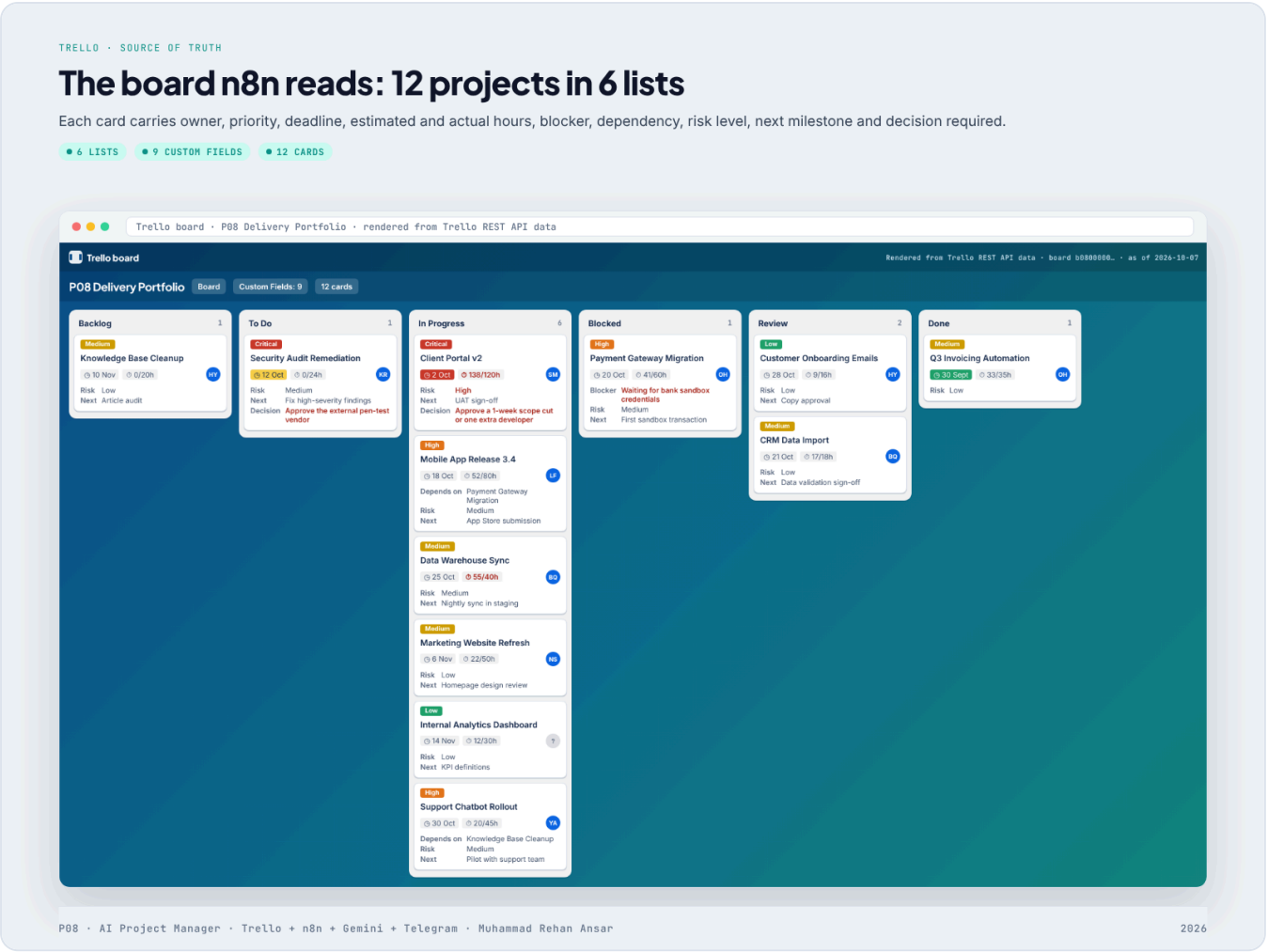
Main path and fallback path



05 • HOW THE AUTOMATION WORKS

1. The schedule (weekdays 08:00) or the Run Now button starts the workflow. A Config node holds the board ID, chat ID, model name and timezone.
2. Three HTTP Request nodes read the board's lists, custom field definitions and open cards from the Trello API (3 retries, 30 s timeout).
3. The Risk Engine node maps each card to a project record and applies the rules. It outputs the result, a rule-based report, and the AI prompt.
4. The Gemini node sends the prompt. If it errors, the flow continues instead of stopping.
5. Select Report validates the AI text and picks the report to send, recording the source and the reason.
6. The Telegram node sends the message to the founder.

12 projects, 6 lists, 9 custom fields



11 active projects: 1 overdue, 1 blocked, 6 at risk, 3 on track

SCENARIO · OVERDUE + AT RISK

Different problems, each with its own reason

One overdue card and five at-risk cards, each caught by a different rule: deadline, hours, missing update, missing owner, or due soon without starting.

OVERDUE HOURS OVER NO UPDATE 15 DAYS MISSING OWNER DUE SOON, NOT STARTED

OVERDUE

Client Portal v2

- Deadline 2026-10-02 passed 5 day(s) ago
- 138h logged vs 120h estimated (+15%)
- Owner marked Risk Level = High

Sara Malik · Critical · due 2026-10-02

AT RISK

Security Audit Remediation

- Due in 5 day(s) but still in "To Do"

Khalid Rahman · Critical · due 2026-10-12

AT RISK

Data Warehouse Sync

- 55h logged vs 40h estimated (+38%)

Bilal Qureshi · Medium · due 2026-10-25

AT RISK

Customer Onboarding Emails

- No update for 15 days

Hana Yusuf · Low · due 2026-10-28

AT RISK

Internal Analytics Dashboard

- Missing: Owner

No owner · Low · due 2026-11-14

AT RISK

Support Chatbot Rollout

- Depends on "Knowledge Base Cleanup" (due 2026-11-10), which finishes after this deadline (2026-10-30)

Yusuf Amin · High · due 2026-10-30

PRECEDENCE	HEALTH	WHEN
1	OVERDUE	Deadline passed and card not Done
2	BLOCKED	Blocker recorded or card in Blocked list
3	AT RISK	Any other signal fires
4	ON TRACK	No signal

SCENARIO · BLOCKER + DEPENDENCY

One blocker, three projects affected

Payment Gateway Migration is blocked. Mobile App Release 3.4 depends on it, so the rules flag it too, even though its own card looks fine.

BLOCKED DEPENDENCY AT RISK DEPENDENCY LATE

Trello card · Payment Gateway Migration

Payment Gateway Migration

in list Blocked - last activity 4 day(s) ago

High

Due 20 Oct

Custom Fields

Owner

Omar Haddad

Priority

High

Risk Level

Medium

Estimated Hours

60

Actual Hours

41

Next Milestone

First sandbox transaction

Blocker

Waiting for bank sandbox credentials

Dependency

Decision Required

Payment Gateway Migration

Blocker: Waiting for bank sandbox credentials

Omar Haddad · High · due 2026-10-20

AT RISK

Mobile App Release 3.4

- Depends on "Payment Gateway Migration", which is BLOCKED

Lina Farouk · High · due 2026-10-18

AT RISK

Support Chatbot Rollout

- Depends on "Knowledge Base Cleanup" (due 2026-11-10), which finishes after this deadline (2026-10-30)

Yusuf Amin · High · due 2026-10-30

HOW THE RULE WORKS

Pass 1 checks each card on its own. Pass 2 looks up every Dependency by card name: if the dependency is OVERDUE or BLOCKED, or will finish after this card's deadline, the dependent card is flagged with the

P08 · AI Project Manager · Case study

5

Gemini gets facts, not decisions

The prompt contains only the computed results as JSON, with fixed instructions: never change or re-judge a status, use only the given facts, name every OVERDUE and BLOCKED project, plain text, at most 170 words.

Three outcomes, all executed in n8n

AI RESPONSE	REPORT SENT	REASON RECORDED BY SELECT REPORT	TELEGRAM
Correct report	AI report	AI report passed validation	ok · #13
Contradicts the rules	Rule report	AI report rejected: AI response calls "Security Audit Remediation" on track, rules say AT RISK	ok · #14
Unavailable (HTTP 503)	Rule report	AI request failed: HTTP 503 — model unavailable	ok · #15

GEMINI · FOUNDER REPORT

AI text accepted only after validation

Select Report received the AI response, checked it against the rule results, and marked it source = ai, "AI report passed validation".

SOURCE = AI

VALIDATED AGAINST RULES

GEMINI-COMPATIBLE TEST ENDPOINT

n8n · Select Report node · output

Execute step

Docs

OUTPUT

SchemaTableJSON

1 item

source	reason	reportText	aiText
ai	AI report passed validation	2 projects need you today; 6 more are at risk. \n \n Act now \n - Client Portal v2 (Sara Malik) is overdue: Deadline 2026-10-02 passed 5 day(s) ago. \n - Payment Gateway Migration (Omar Haddad) is blocked: Blocker: Waiting for bank sandbox credentials. \n \n Watch \n - Security Audit Remediation: Due in 5 day(s) but still in "To Do". \n - Mobile App Release 3.4: Depends on "Payment Gateway Migration", which is BLOCKED. \n - Support Chatbot Rollout: Depends on "Knowledge Base Cleanup" (due 2026-11-10), which finishes after this deadline (2026-10-30). \n - Data Warehouse Sync: 55h logged vs 40h estimated (+38%). \n - Customer Onboarding Emails: No update for 15 days. \n - Internal Analytics Dashboard: Missing: Owner.	2 projects need you today; 6 more are at risk. \n \n Act now \n - Client Portal v2 (Sara Malik) is overdue: Deadline 2026-10-02 passed 5 day(s) ago. \n - Payment Gateway Migration (Omar Haddad) is blocked: Blocker: Waiting for bank sandbox credentials. \n \n Watch \n - Security Audit Remediation: Due in 5 day(s) but still in "To Do". \n - Mobile App Release 3.4: Depends on "Payment Gateway Migration", which is BLOCKED. \n - Support Chatbot Rollout: Depends on "Knowledge Base Cleanup" (due 2026-11-10), which finishes after this deadline (2026-10-30). \n - Data Warehouse Sync: 55h logged vs 40h estimated (+38%). \n - Customer Onboarding Emails: No update for 15 days. \n - Internal Analytics Dashboard: Missing: Owner.

AI REPORT TEXT (AS SENT)

2 projects need you today; 6 more are at risk.

Act now

- Client Portal v2 (Sara Malik) is overdue: Deadline 2026-10-02 passed 5 day(s) ago.

- Payment Gateway Migration (Omar Haddad) is blocked: Blocker: Waiting for bank sandbox credentials.

Watch

- Security Audit Remediation: Due in 5 day(s) but still in "To Do".

- Mobile App Release 3.4: Depends on "Payment Gateway Migration", which is BLOCKED.

- Support Chatbot Rollout: Depends on "Knowledge Base Cleanup" (due 2026-11-10), which finishes after this deadline (2026-10-30).

- Data Warehouse Sync: 55h logged vs 40h estimated (+38%).

- Customer Onboarding Emails: No update for 15 days.

- Internal Analytics Dashboard: Missing: Owner.

Decisions for you

- Client Portal v2: Approve a 1-week scope cut or one extra developer.

- Security Audit Remediation: Approve the external pen-test vendor.

On track: CRM Data Import, Knowledge Base Cleanup, Marketing Website Refresh.

What the founder receives

TELEGRAM • NOTIFICATION

What the founder receives

Sent by the n8n Telegram node. The Telegram Bot API answered ok = true (right: the Bot API response from the recorded editor run). Chat and bot IDs are masked.

• DELIVERED • HTML FORMAT • RETRY 3*

P08 Project Health — 2026-10-07

🔴 Overdue 1 🚫 Blocked 1 🟡 At risk 6 🟢 On track 3

Status decided by rules - report by Gemini-compatible test endpoint (validated)

2 projects need you today; 6 more are at risk.

Act now

- Client Portal v2 (Sara Malik) is overdue: Deadline 2026-10-02 passed 5 day(s) ago.
- Payment Gateway Migration (Omar Haddad) is blocked: Blocker: Waiting for bank sandbox credentials.

Watch

- Security Audit Remediation: Due in 5 day(s) but still in "To Do".
- Mobile App Release 3.4: Depends on "Payment Gateway Migration", which is BLOCKED.
- Support Chatbot Rollout: Depends on "Knowledge Base Cleanup" (due 2026-11-10), which finishes after this deadline (2026-10-30).
- Data Warehouse Sync: 55h logged vs 40h estimated (+38%).
- Customer Onboarding Emails: No update for 15 days.
- Internal Analytics Dashboard: Missing: Owner.

Decisions for you

- Client Portal v2: Approve a 1-week scope cut or one extra developer.
- Security Audit Remediation: Approve the external pen-test vendor.

On track: CRM Data Import, Knowledge Base Cleanup, Marketing Website Refresh.

22:54 UTC ✓✓

n8n • Telegram node • Bot API response

OUTPUT

Schema Table JSON

This node will automatically retry if it fails

1 item

ok	result
true	<pre> message_id : 16 from id : ***** is_bot : true first_name : AI Project Manager username : rehan_ai_project_manager_bot chat id : ***** first_name : Rehan last_name : khan type : private date : 1791413705 text : P08 Project Health — 2026-10-07\ 1 🔴 Overdue 1 🚫 Blocked 1 🟡 At risk 6 🟢 On track 3\ Status decided by rules - report by Gemini-compatible test endpoint (validated)\ 2 projects need you today; 6 more are at risk.\ Act now\ - Client Portal v2 (Sara Malik) is overdue: Deadline 2026-10-02 passed 5 day(s) ago.\ - Payment Gateway Migration (Omar Haddad) is blocked: Blocker: Waiting for bank sandbox credentials.\ Watch\ - Security Audit Remediation: Due in 5 day(s) but still in "To Do".\ - Mobile App Release 3.4: Depends on "Payment Gateway Migration", which is BLOCKED.\ - Support Chatbot Rollout: Depends on "Knowledge Base Cleanup" (due 2026-11-10), which finishes after this deadline (2026-10-30).\ - Data Warehouse Sync: 55h logged vs 40h estimated (+38%).\ - Customer Onboarding Emails: No update for 15 days.\ - Internal Analytics Dashboard: Missing: Owner.\ Decisions for you\ - Client Portal v2: Approve a 1-week scope cut or one extra developer.\ - Security Audit Remediation: Approve the external pen-test vendor.\ On track: CRM Data Import, Knowledge Base Cleanup, Marketing Website Refresh. </pre>

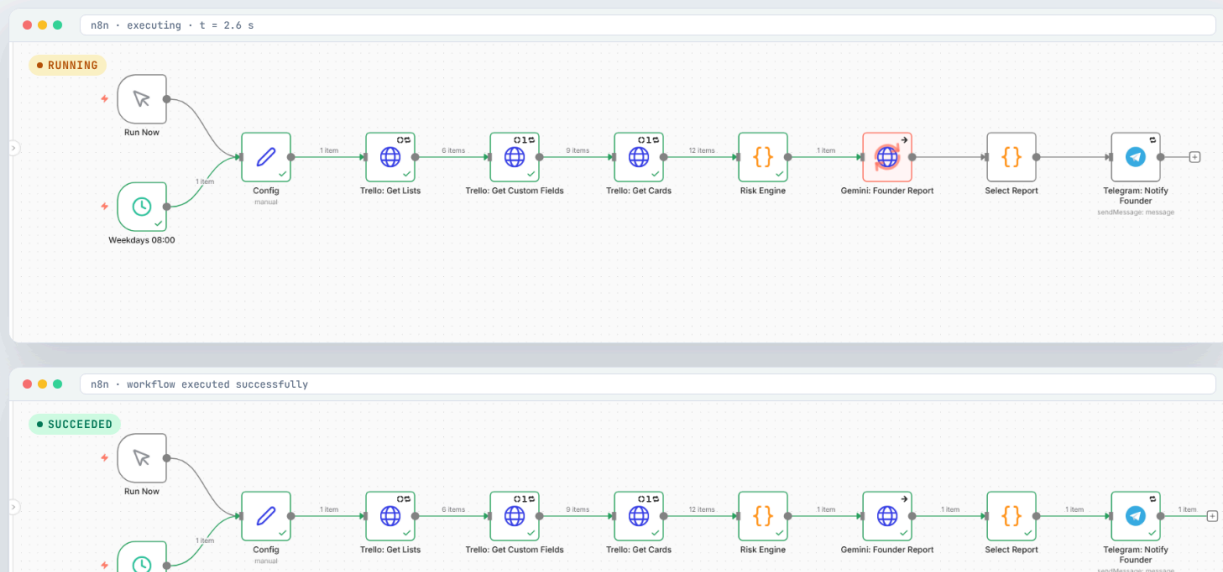
Captured during a real n8n execution

N8N • LIVE EXECUTION

Captured while running: 4.3 s from click to finish

Top: 2.6 s after clicking Execute — Trello and Risk Engine done, Gemini running. Bottom: every node green, item counts on each connection.

• RUNNING • SUCCEEDED • 34/34 TESTS



34 of 34 automated tests pass · 12 acceptance scenarios

#	SCENARIO	EXPECTED AND VERIFIED	WHERE
S1	All projects on track	No flags, no "act now" section	Unit
S2	Overdue project	OVERDUE, days late in reason	Unit + n8n
S3	Blocked project	BLOCKED, blocker text in reason	Unit + n8n
S4	Dependency problem	Blocked / late / unknown dependency flagged	Unit + n8n
S5	Hours exceeded	+38% → high severity	Unit + n8n
S6	Multiple risks on one card	6 flags, OVERDUE wins by precedence	Unit
S7	Missing update	15 days → high severity	Unit + n8n
S8	Missing required data	Missing owner listed	Unit + n8n
S9	Gemini unavailable	HTTP 503 → rule report sent	Unit + n8n
S10	Gemini correct	AI report accepted	Unit + n8n
S11	Gemini contradicts rules	Rejected → rule report sent	Unit + n8n
S12	Telegram notification	Delivered, Bot API ok = true	Unit + n8n

What works today

- Importable n8n workflow, executed end to end in n8n 1.123 with every node succeeding.
- Correct health and a reason for all 12 projects; same answer on every run.
- AI report used only when it agrees with the rules; rule report otherwise. All three paths delivered to Telegram.
- Credentials stored only in n8n; none in code, documents or git.

TEST ENVIRONMENT

The n8n runs used the production workflow file with a local Trello API stand-in serving the board data, and a Gemini-API-compatible test endpoint for the AI step. Telegram delivery was live. Connecting a live Trello board and Gemini key needs only the three n8n credentials described in the HOW-TO.